



اوتنورسيتي مليميا فهغ السلطان عبد الله
UNIVERSITI MALAYSIA PAHANG
AL-SULTAN ABDULLAH

BCS2313

ARTIFICIAL INTELLIGENCE TECHNIQUES

FUZZY LOGIC PROJECT

SESSION I 2025/2026

LECTUER 'S NAME: Ts. Dr. NUR SHAZWANI KAMARUDIN

SECTION: 01A

GROUP MEMBERS AND MATRIC NO:

LUQMAN SHAHIDIN BIN MUSTAFA (CD23043)

ZULAKMAL DANISH BIN ZULFADZLI (CD23054)

SAKTHIVELL A/L PALANI (CD23047)

MUHAMMAD SYAHMI SOLIHIN BIN MUSTAFA (CB20086)

DATASET NAME: ROAD TRAFFIC ACCIDENT

PROGRAMMING LANGUAGE: PYTHON

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1.0 DATASET INFORMATION

1.1 OBJECTIVE

1.2 ATTRIBUTES

1.2.1 INPUT VARIABLES

1.2.2 OUTPUT VARIABLE (TARGET LABEL)

1.3 TOTAL DATASET

1.4 SCREENSHOT SAMPLE LIST OF THE DATA

1.5 JOURNAL PAPERS

First Journal

Fuzzy Logic-Based Accident Detection System for Vehicular Ad Hoc Networks: A Prototype Implementation

Authors: Urmila Bhanja, Anita Mohanty, Sudipta Mahapatra

Year: 2024

Reference Link: <https://ieeexplore.ieee.org/abstract/document/8951422>

Summary

This research proposes an intelligent vehicle accident detection system with the Vehicular Ad Hoc Network (VANET) model that is supported with a hardware prototype and not a static data set. Dataset: The study did not utilize an existing dataset, but instead, it created real-time data through a prototype that had sensors. The input parameters were vehicle parameters (speed, orientation with gyroscope, and impact force) and driver condition parameters (heart rate, blood pressure, muscle strain, voice intensity and eye aspect ratio).

Rules: The system used a Mamdani Fuzzy Logic Controller (MFLC) which had individual rule sets in the modules. As an example, one of the rules in the driver module is: IF Blood Pressure is Low and Pulse Rate is Low and Muscle Strain is Low then Driver Condition is Severe. An example of a vehicle rule is: "IF Speed is High AND Gyroscope is Normal AND Force is High THEN Vehicle Condition is Tilt).

Results: The hardware prototype showed high levels of classification with a classification accuracy of 94.26 percent during a six-month period in which it monitored 720 events. The system registered 76 true positive accident detections and only 19 false positives and this showed that the fuzzy logic system was effective in reducing false alarm relative to the single parameter systems.

Second Journal

Fuzzy Logic based Recommendation System for Road Accident Data Analysis

Authors: S. Saranyadevi, R. Murugeswari, S. Bathrinath

Year: 2025

Reference Link: <https://link.springer.com/article/10.1007/s11277-024-11500-w>

Summary

This paper presents a novel data mining method combining fuzzy logic with formal grammar (Context-Free Grammar) to analyze road accident risks and provide safety recommendations.

Dataset: The dataset used in the study is a historical traffic road accident dataset in the year 2015 provided by the Government of India. The data was 2000 records of 7 attributes where the type of vehicles, age of the person, gender, weather conditions, time of accident, location and nature of accident are considered.

Rules: The algorithm used by the researchers was a Grammatical Fuzzy Rule Mining, which is a type of algorithm in which Context-Free Grammar (CFG) would generate fuzzy production rules without any human intervention. In contrast to the old-fashioned way of defining rules manually, this system employed the grammar to assign membership values and frame rules to determine associations, like the one that connects Two-wheelers and the age group under 18.

Results: The test Grammatical Fuzzy Rule Mining algorithm is better than conventional fuzzy rule mining algorithms. The experimental outcome depicted an accuracy of about 80, precision of about 0.85 and a recall of about 0.9 which is better than the comparative traditional fuzzy approach. The analysis revealed that the most prone to accidents were the two-wheeler and lower age group, (under 18 years of age).

Third Journal

Fuzzy Logic Model for Assessing Accident Proneness Based on Passenger Vehicle Speed in Real and Virtual Traffic Conditions

Authors: Nenad Marković, Tijana Ivanišević, Svetlana Čičević, Aleksandar Trifunović

Year: 2024

Reference Link: <https://www.mdpi.com/2227-7390/12/3/421>

Summary

In this research, fuzzy logic models were developed in predicting the proneness of a driver to have an accident depending on the perception of the driver of the vehicle speed in the real-world and the simulated situation.

Dataset: The scientists gathered the empirical data of 87 young drivers (mean age 21.54). The participants measured the speed of vehicles in two settings: Real Traffic Conditions (RTC) and a Driving Simulator (DS) of 30, 50, and 70 km/h. The perceived speeds were the input variables, and the number of self-reported road crashes was the output variable.

Rules: A Mamdani Fuzzy Inference System had been developed on Matlab with 10 rules relating errors in perception of speed to the risk of accidents. An example was as follows: Rule 1: IF (30km/h is VPLSE) and (50km/h is VPLSE) and (70km/h is VPLSE) THEN (Roadtrafficaccidents is VLNTA) (where VPLSE is Very Poor Low Speed Estimation and VLNTA is Very Large Number of Traffic Accidents).

Results: The research result showed statistically significant differences in the perception of speed in the real world and the virtual world where drivers tended to underestimate the speeds in the simulator. Cumulative Deviation (CD) was used to validate the fuzzy logic model and t-tests were done to demonstrate that there was no statistically significant difference in the model predictions and the actual history of accidents of the participants, and this confirmed the accuracy of the model in its ability to predict accidents proneness.

2.0 FUZZY LOGIC FRAMEWORK

1. DATA PREPROCESSING

2.FUZZIFICATION

3.FUZZY RULE BASE

4.INFERENCE ENGINE

5.DEFUZZIFICATION

6.OUTPUT GENERATION AND EVALUATION

2.1 FUZZY LOGIC MEMBERSHIP

3.0 RESULTS

4.0 CONCLUSION

SOURCE CODE

REFERENCES